

October 9th, 2025

SMART KUBERNETES

AI-POWERED CLUSTER OPTIMIZATION WITHOUT BREAKING THE BANK

Hello



Kunal Das

Developer Advocate APAC, CAST AI

Organizer of
CNCf Kolkata,
Cloud Computing Circle,
Hashicorp User Group Bangalore

7x Azure, 1x Hashicorp Certified, FinOps Certified
Engineer



Kubernetes is easy
Isn't it?



PROJECTS AND PRODUCTS

Cloud Native Ecosystem RELM, K8s, Helm, Istio, Prometheus, Grafana, etc.	Cloud Native Ecosystem KV, V, etc.	Cloud Native Ecosystem Flux, Argo, etc.	Cloud Native Ecosystem etc.
Cloud Native Ecosystem etc.	Cloud Native Ecosystem etc.	Cloud Native Ecosystem etc.	Cloud Native Ecosystem etc.
Cloud Native Ecosystem etc.	Cloud Native Ecosystem etc.	Cloud Native Ecosystem etc.	Cloud Native Ecosystem etc.
Cloud Native Ecosystem etc.	Cloud Native Ecosystem etc.	Cloud Native Ecosystem etc.	Cloud Native Ecosystem etc.
Cloud Native Ecosystem etc.	Cloud Native Ecosystem etc.	Cloud Native Ecosystem etc.	Cloud Native Ecosystem etc.
Cloud Native Ecosystem etc.	Cloud Native Ecosystem etc.	Cloud Native Ecosystem etc.	Cloud Native Ecosystem etc.

How to handle the zoo?

Cli ninja skills



Automation



Automation kills my job 🤖

NO WAY..!

Automation is the key ingredient

These culprits exist because managing cloud-native infrastructure **manually** is nearly impossible.

Automation tools:

- Select the type, size, and number of compute instances your K8s workloads actually need
- Bin-pack pods into nodes to maximize resource utilization
- Manage the entire Spot instance lifecycle,
- Set the right requests to fight cloud waste while maintaining performance (rightsize workloads)

**Kubernetes is efficient by default,
no?**

Key findings

CPU and memory utilization

10%

CPU utilization across clusters is **worse** this year

AVERAGE CPU UTILIZATION

23%

AVERAGE MEMORY UTILIZATION

Surprisingly, the average CPU utilization across clusters is **worse** this year than last year by a lot more than we expected: now at 10% (23% worse than in 2024), while average memory utilization was marginally better at 23%, indicating no significant year-over-year improvement in resource efficiency across cloud platforms compared to our previous report from 2024.

[Link](#)

Let's talk about infrastructure

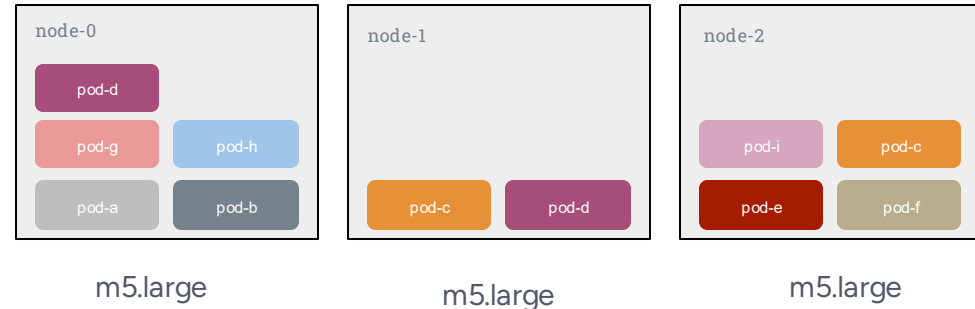
Autoscaling with efficient compaction

Kubernetes distribute Workloads on nodes

Distribution is adjusted based on need,
configuration, workload

Only nodes cost money! (ok also traffic and
control plane)

Cost saving is not so complicated!



Autoscaling with efficient compaction

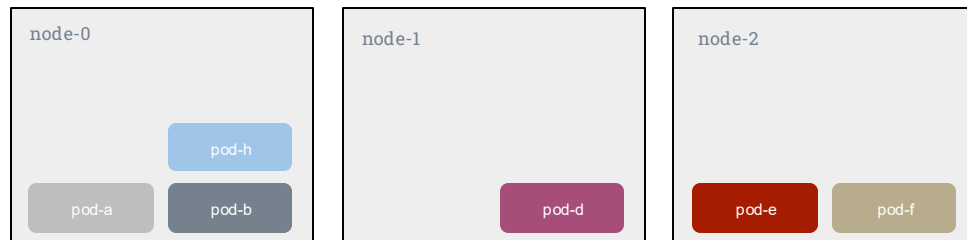
Internal Scheduling and scaling is cost neutral

Resource sharing makes Kubernetes efficient

Underutilized nodes are expensive

Make use of the elasticity of the cloud

Make sure you have a bin-packing solution



Autoscaling with efficient compaction

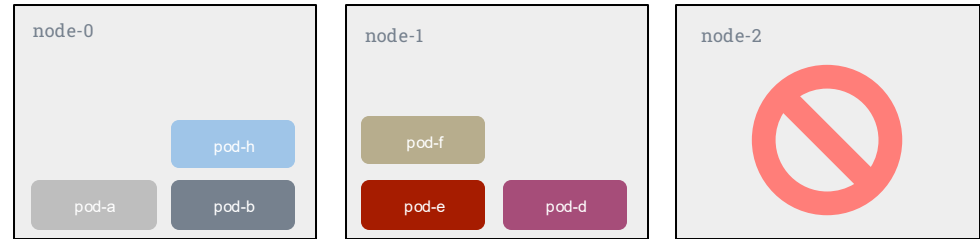
Internal Scheduling and scaling is cost neutral

Resource sharing makes Kubernetes efficient

Underutilized nodes are expensive

Make use of the elasticity of the cloud

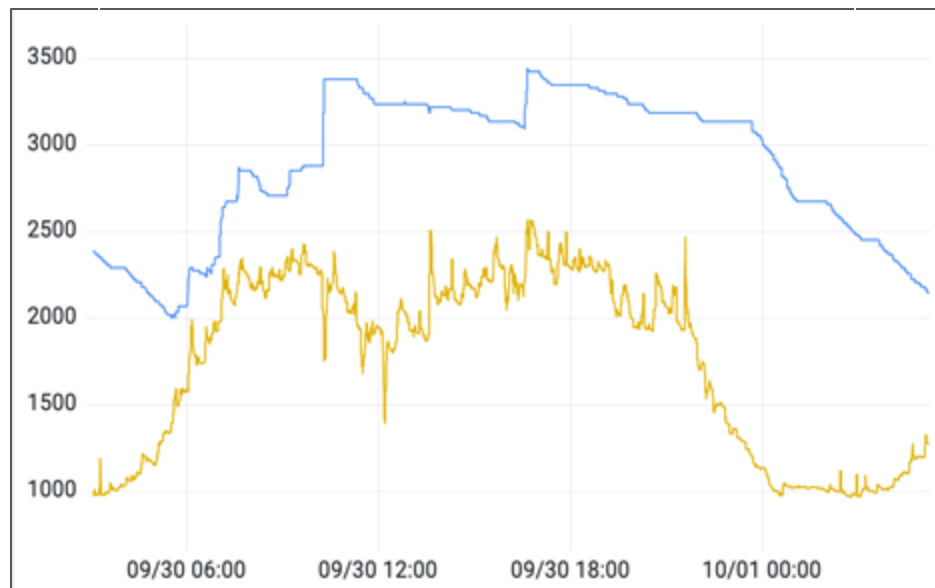
Make sure you have a bin-packing solution



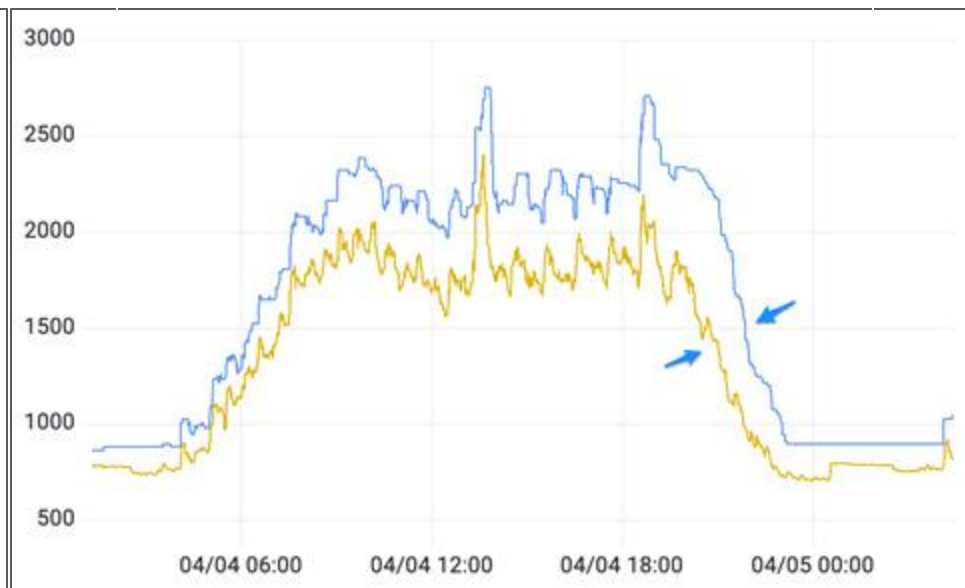
CAST AI autoscaler

Reducing the gap between provisioned and used resources

Before optimization



After optimization

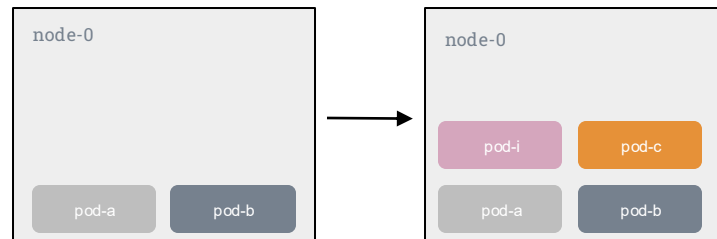


What about workload?

HPA and VPA

Horizontal Pod Autoscaler

Adjust number of pods for load (CPU, requests)

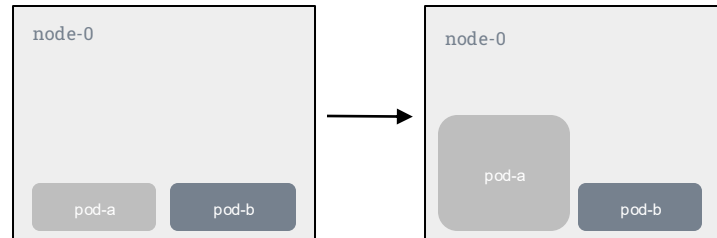


Vertical Pod Autoscaler

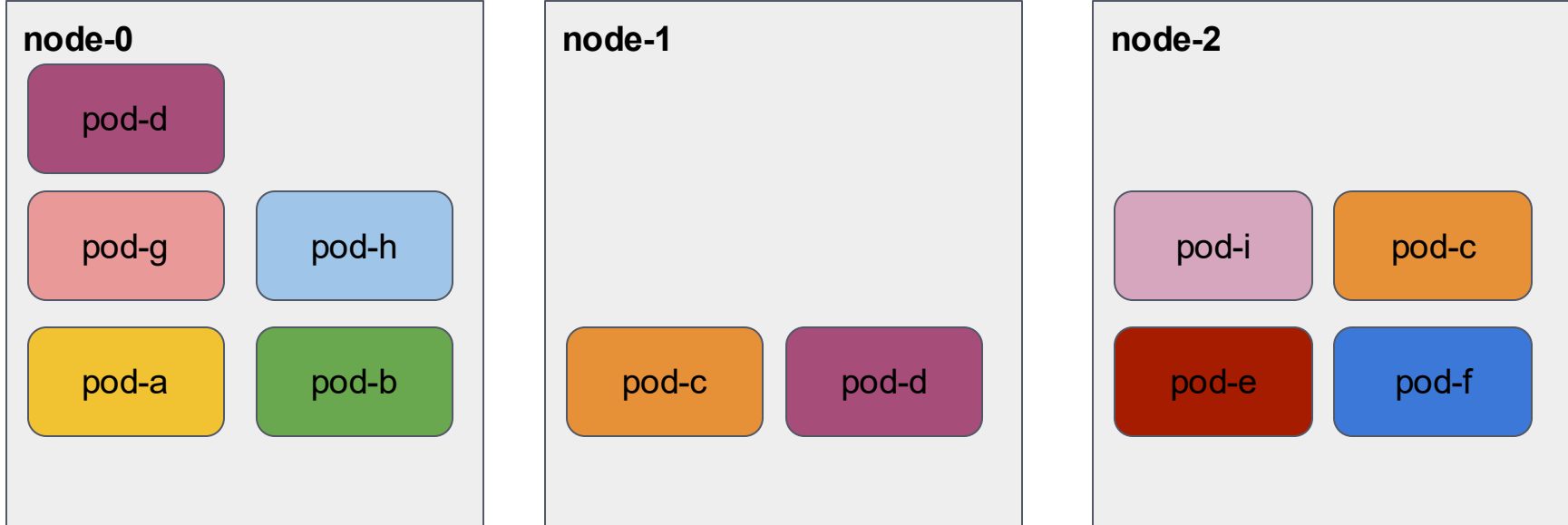
Requirement for efficient nodes scaling

Adjust pod resource request

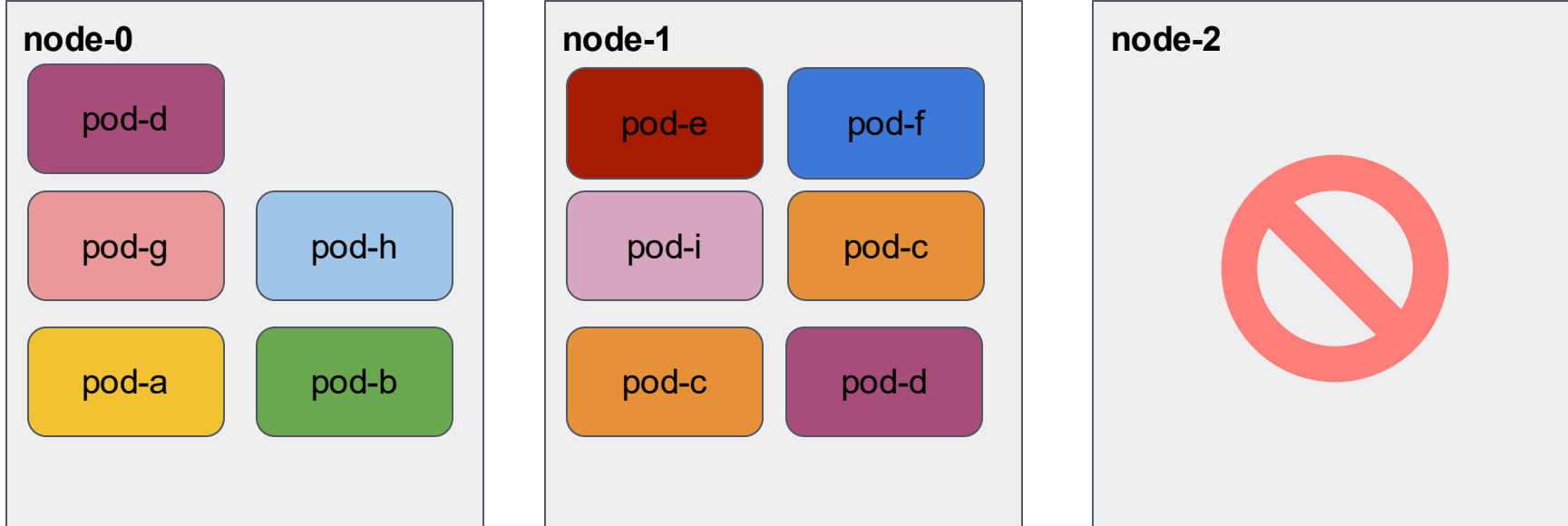
Dangerous side effects (pod too big to schedule)



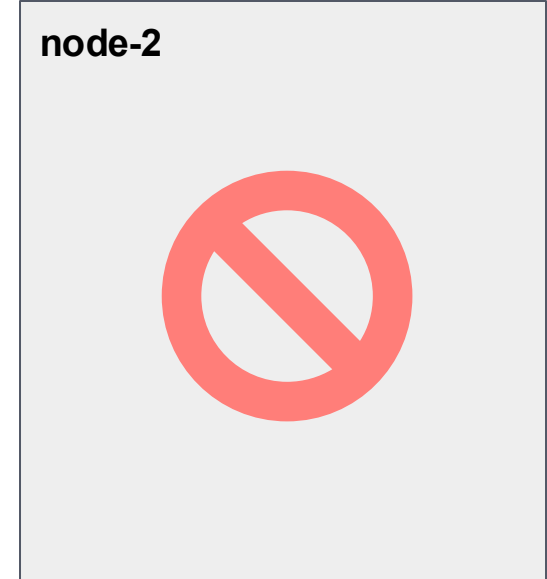
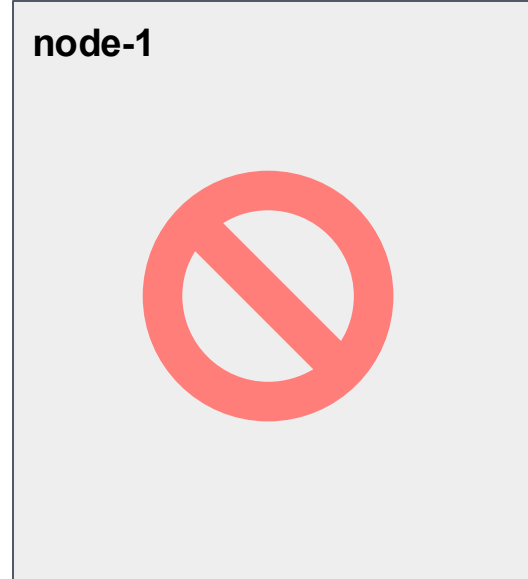
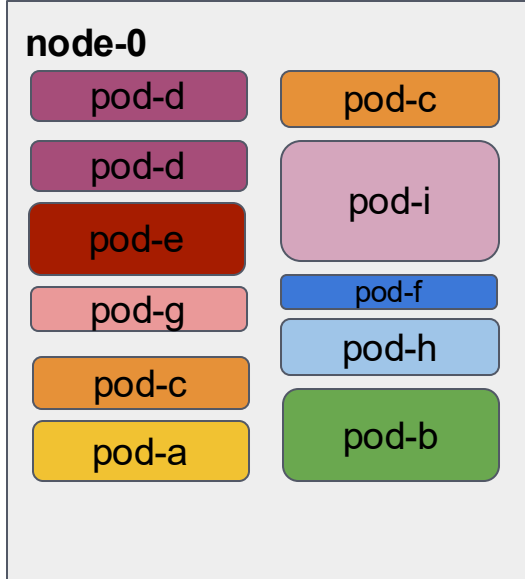
Combining all of it - Before



Combining all of it - Before



Combining all of it - Before



Cast AI works with your favourite tools, platforms, and technologies.



So..... everything is awesome?

The Problem

Kubernetes supports stateless workloads well, but what about stateful workloads?

1

Stateful workloads can't be stopped and restarted without data loss or interruption of service

2

Underutilized nodes that have stateful workloads can't be consolidated and rotated without interruption of service

3

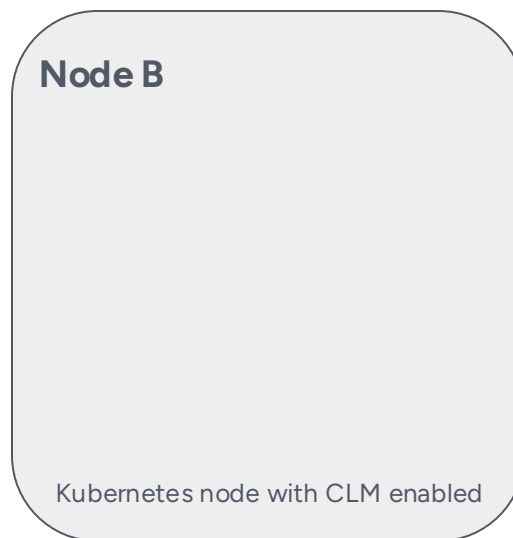
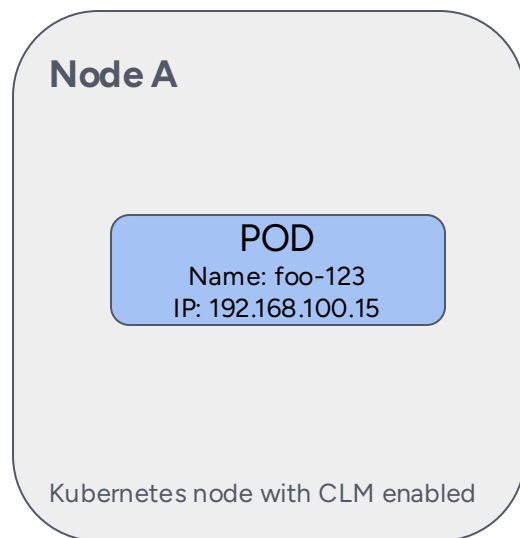
Engineers get caught in a tug-of-war between optimizing cloud costs, node security and reliability

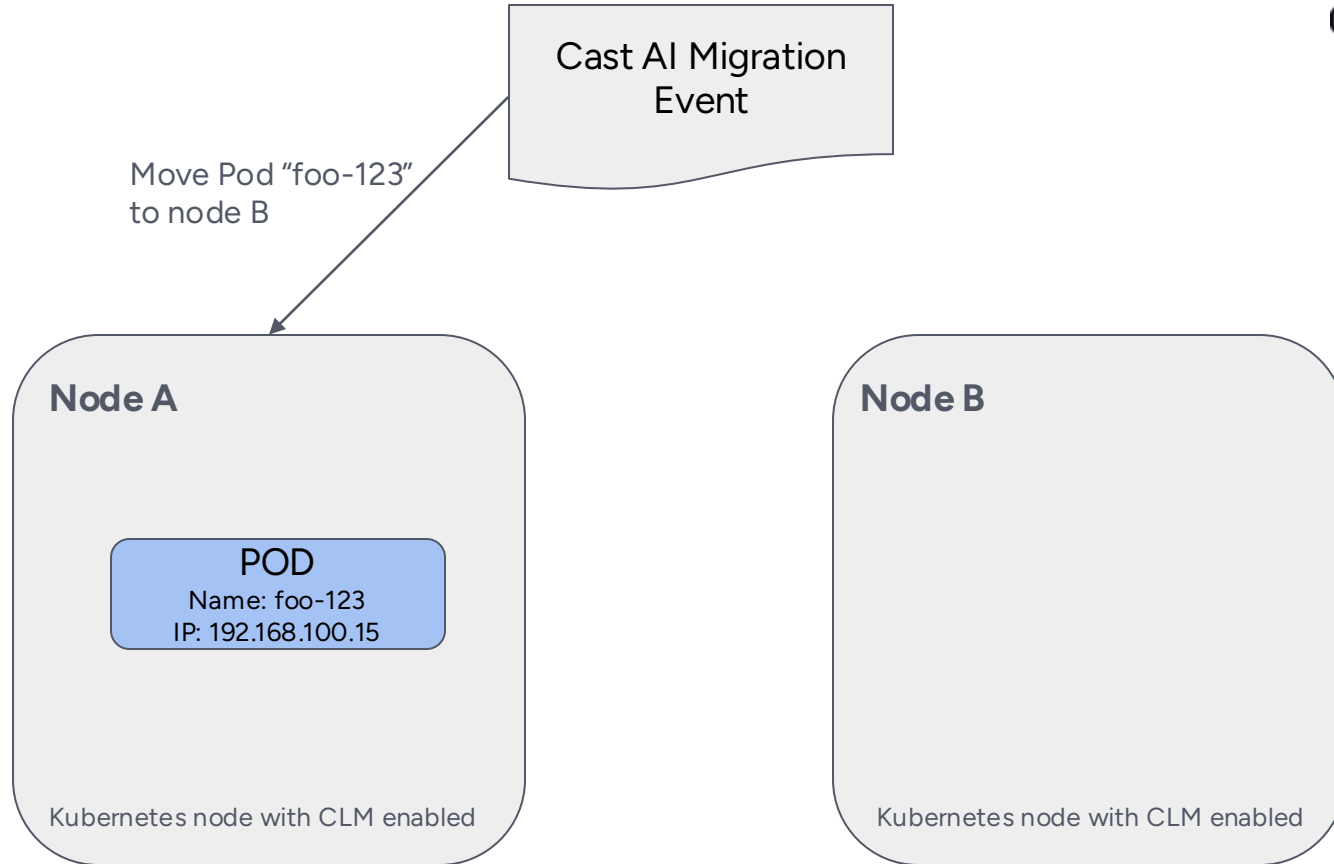
What does “Container Live Migration” means?

The ability to move containers (pods) in the **SAME** cluster
between nodes without interruption and restarts.

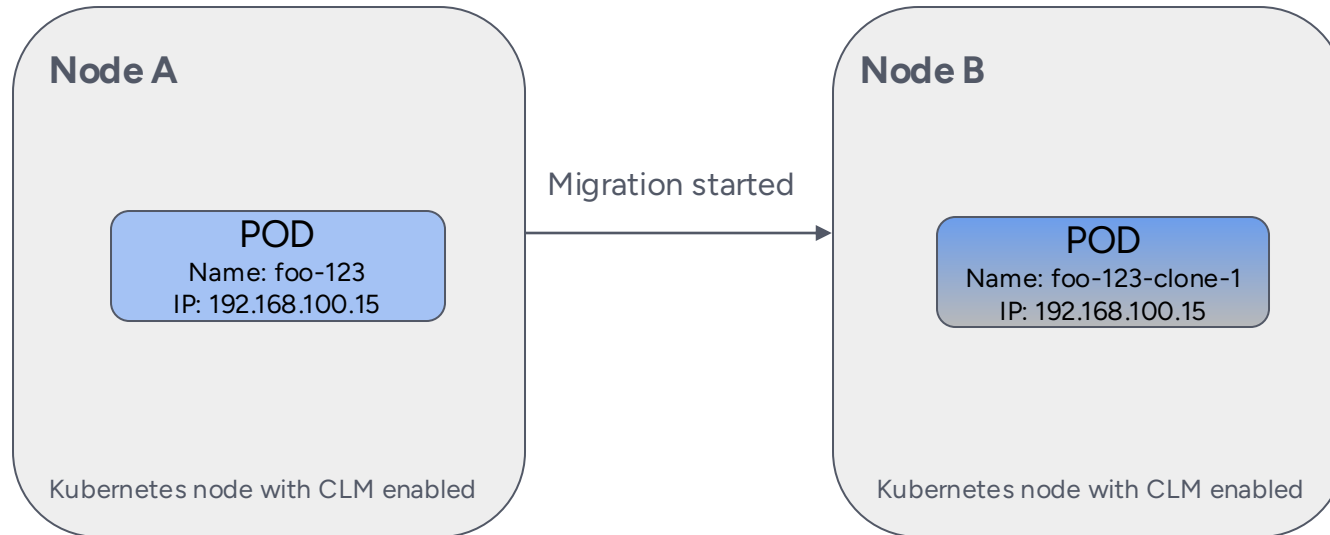


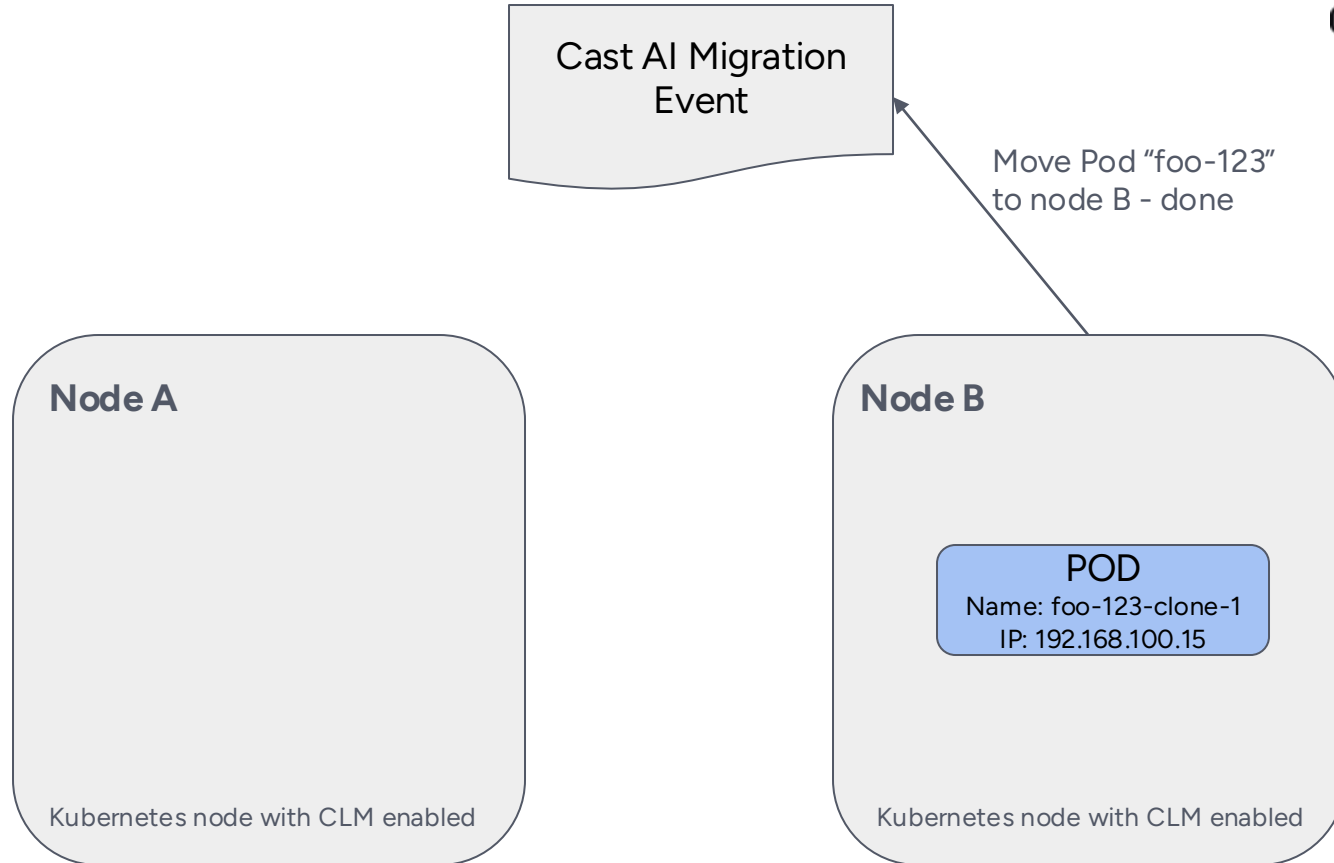
How does it work?





Cast AI Migration Event





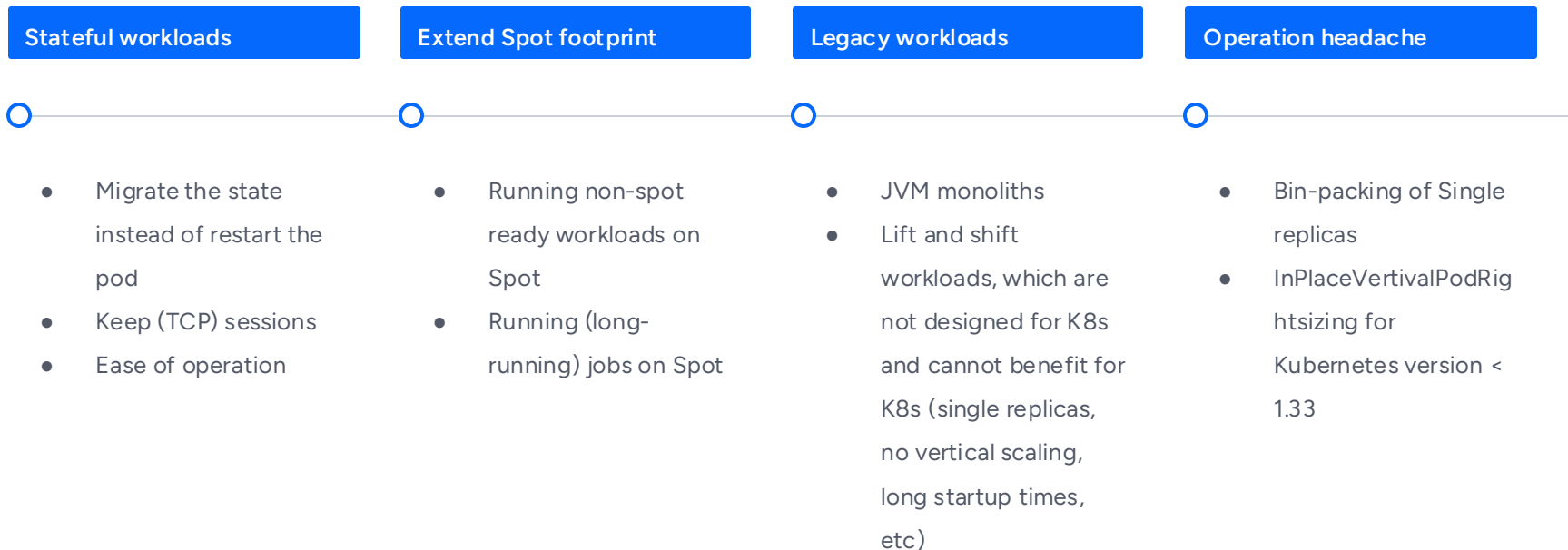
No interruption, no downtime, no restart



But wait.... Why do i need this in
Kubernetes environments?



Some of the use cases



When is it GA?

AWS EKS

NOW 🙌



GCP GKE

Q4 / 2025 😊



Azure AKS

2026 😞



**Let's live migrate a Minecraft server during
we're playing**

THANK YOU



PING ME ANYTIME TO
CHAT ABOUT
ANYTHING

